

- Consider a robot with two powered spherical wheels and a passive castor wheel (no power). Derive the feedforward kinematic solution $\dot{\xi}_I$ for this robot, that is, derive $\dot{\xi}_I$ of the form of the solution like Eq. 3.30 in Page 74 or Eq. 3.33 in Page 76 of the textbook, i.e.

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} 1 & 0 & l \\ 1 & 0 & -l \\ 0 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} J_2 \dot{\phi} \\ 0 \end{bmatrix} = R(\theta)^{-1} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2l} & -\frac{1}{2l} & 0 \end{bmatrix} \begin{bmatrix} J_2 \dot{\phi} \\ 0 \end{bmatrix} \quad \dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3l} & -\frac{1}{3l} & -\frac{1}{3l} \end{bmatrix} J_2 \dot{\phi}$$

- See also in Problem 2 of Chapter 3.7 in the textbook.